

(A) Test Patterns & Parameters

This appendix describes the Threshold and Suprathreshold test parameters and test patterns.

Threshold Test Patterns

Default pattern is shown in bold.

Test Pattern	Extent of Visual Field Tested / Number of Points Tested	Application
Central 30-2	30 degrees/76 point grid	Glaucoma, retinal, neurological, general
Central 24-2	24 degrees/54 point grid	Glaucoma, general, neurological
Central 10-2	10 degrees/68 point grid	Macula, retinal, neurological, advanced glaucoma
Central 24-2C	24 degrees/64 point grid	Glaucoma, general, neurological
Peripheral 60-4	30 to 60 degrees/60 points	Retinal, glaucoma
Macula	5 degrees/16 points, 2 degrees spacing	Macula
Nasal Step	50 degrees/14 points	Glaucoma

Table A-1

Threshold Test Pattern Diagrams

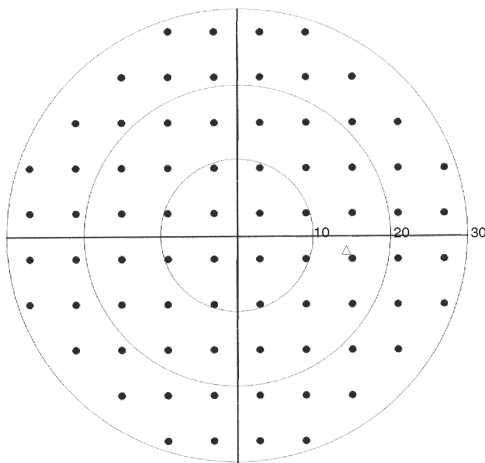


Figure A-1 Central 30-2 Test Pattern, Right Eye

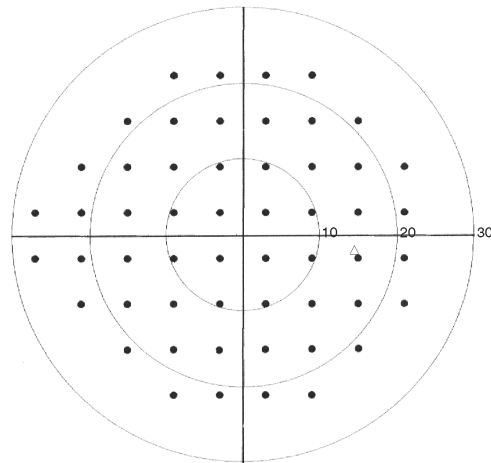
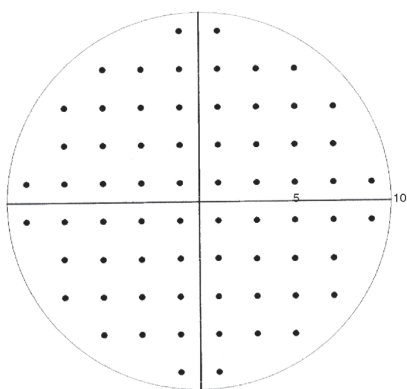
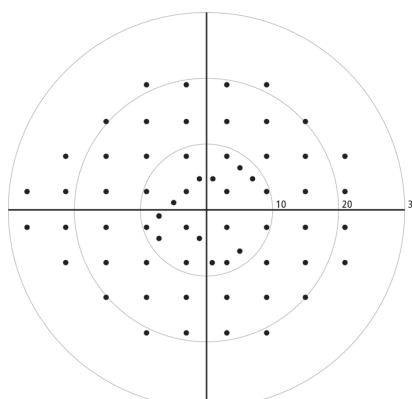
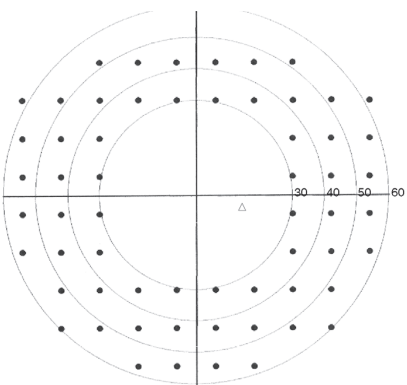
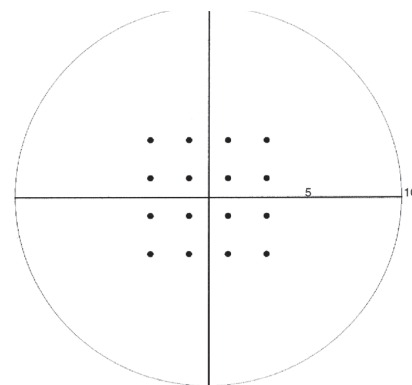
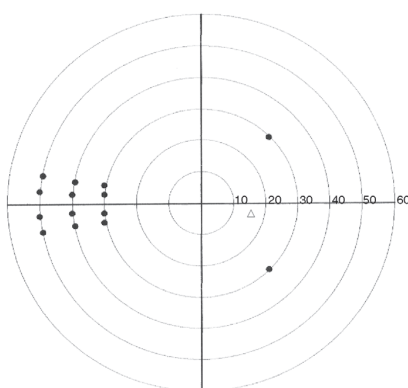


Figure A-2 Central 24-2 Test Pattern, Right Eye

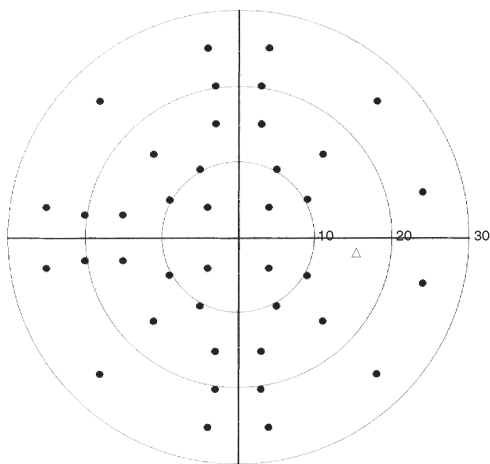
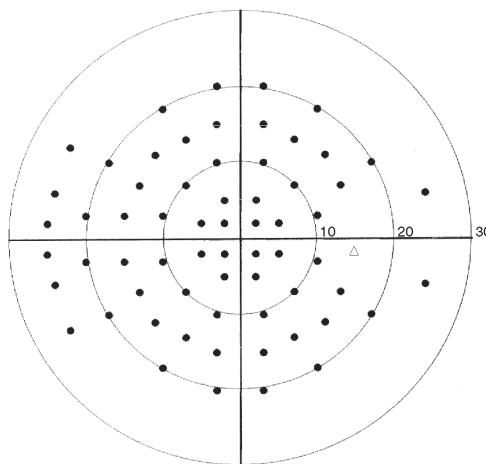
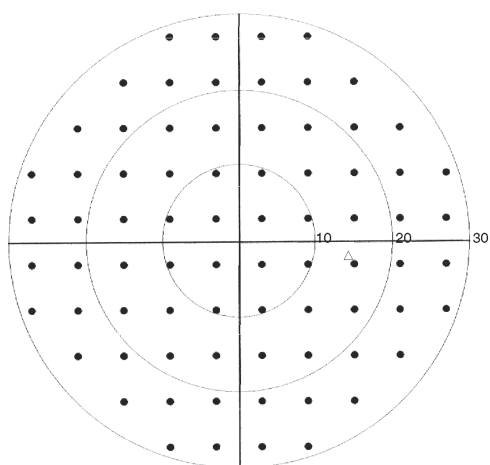
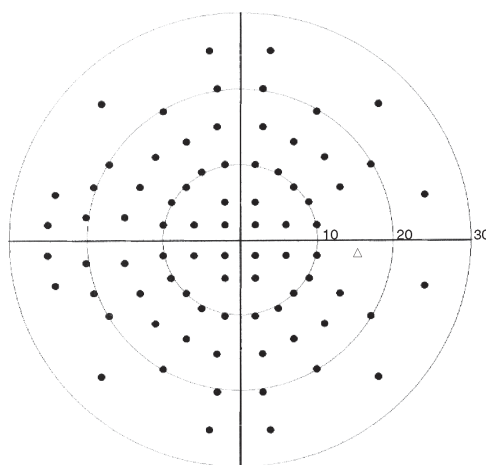
*Figure A-3 Central 10-2 Test Pattern**Figure A-4 Central 24-2C Test Pattern, Right Eye**Figure A-5 Peripheral 60-4 Test Pattern, Right Eye**Figure A-6 Macula Test Pattern**Figure A-7 Nasal Step Test Pattern, Right Eye*

Suprathreshold Test Patterns

Default pattern is shown in bold.

Test Pattern	Extent of Visual Field Tested / Number of Points Tested	Application
Central 40 Point	30 degrees/40 points	General screening
Central 64 Point	30 degrees/64 points	General, glaucoma, neurological
Central 76 Point	30 degrees/76 points	General, glaucoma, neurological
Central 80 Point	30 degrees/ 80 points	General Screening
Armaly Central	30 degrees/84 points	Glaucoma
Nasal Step	50 degrees/14 points	Glaucoma
Peripheral 60 Point	30 to 60 degrees/60 points	General, neurological with central exam, retinal, glaucoma
Full Field 81 Point	55 degrees/81 points	General, retinal, glaucoma, neurological
Full Field 120 Point	55 degrees/120 points	General, retinal, glaucoma, neurological
Full Field 135 Point	87 degrees/135 points 87 degrees temporally	Full Field Screening
Full Field 246 Point	60 degrees/246 points	Full Field Screening
Armaly Full Field	50 degrees/98 points	Glaucoma
Superior 36 Point	60 degrees, superior hemifield/36 points	Superior Field Screening, Ptosis
Superior 64 Point	60 degrees, superior hemifield/64 points	Superior Field Screening, Ptosis
Esterman Monocular	75 degrees temporal 60 degrees nasal/100 points	Functional disability
Esterman Binocular	150 degrees bitemporal/ 120 points	Functional disability
Blindengeldgutachten	50 degrees/68 points	German Disability Test
Fuehrerscheingutachten	80 degrees temporal, 60 degrees nasal/105 points	German Drivers License Test

Table A-2

Suprathreshold Test Pattern Diagrams*Figure A-8 Central 40-Point Test Pattern, Right Eye**Figure A-9 Central 64-Point Test Pattern, Right Eye**Figure A-10 Central 76-Point Test Pattern, Right Eye*
(This is the same pattern as the 30-2 Threshold test)*Figure A-11 Central 80-Point Test Pattern, Right Eye*

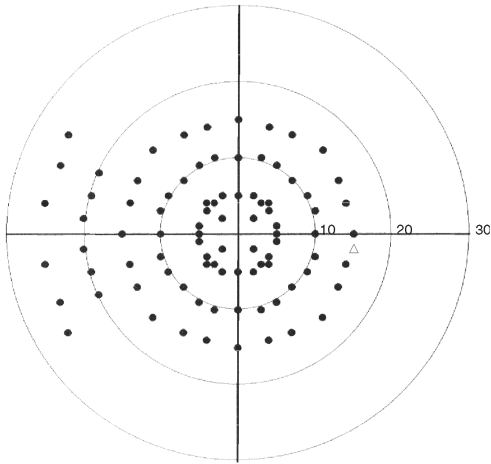


Figure A-12 Central Armaly Test Pattern, Right Eye

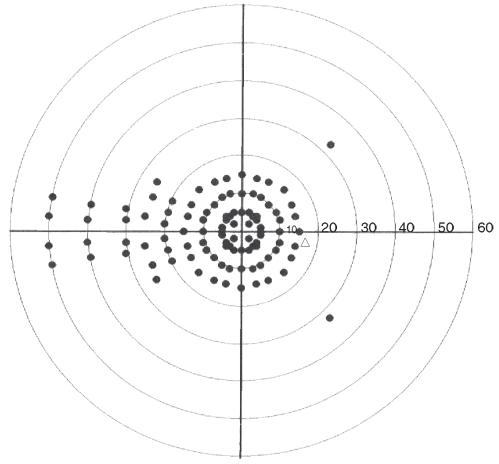


Figure A-13 Full Field Armaly Test Pattern, Right Eye

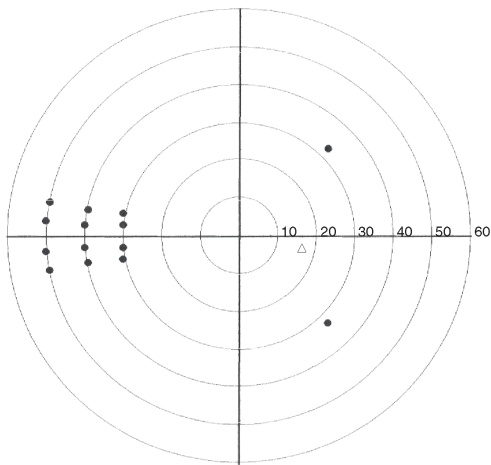


Figure A-14 Nasal Step Test Pattern, Right Eye

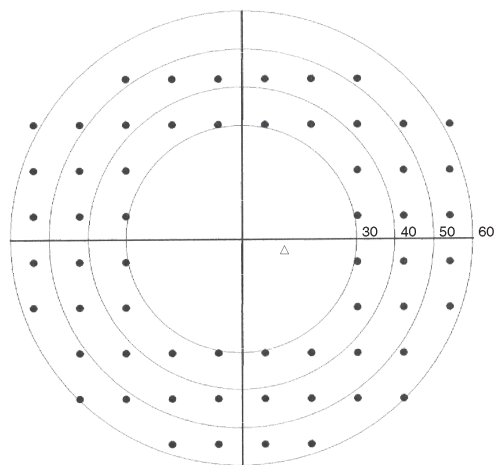
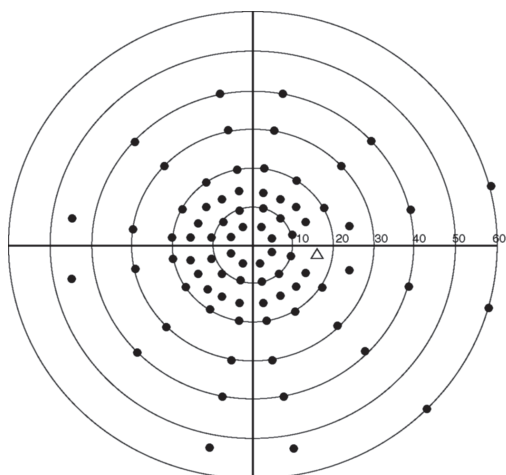
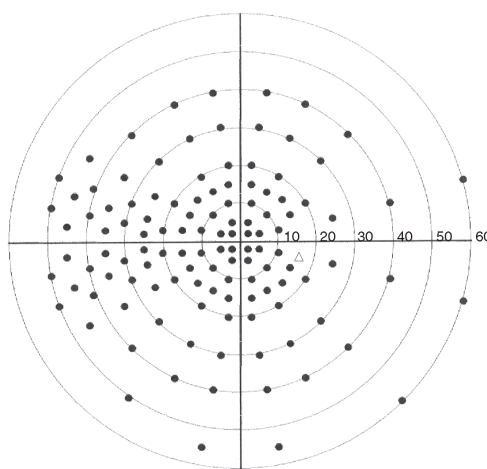
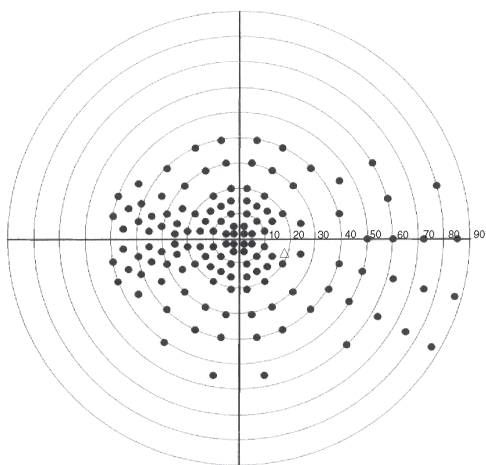
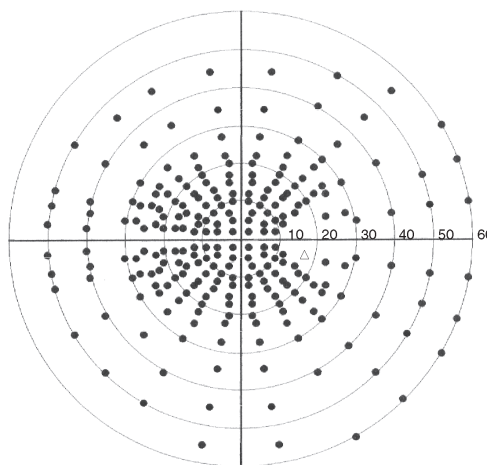


Figure A-15 Peripheral 60 Test Pattern, Right Eye

*Figure A-16 Full Field 81 Test Pattern, Right Eye**Figure A-17 Full Field 120 Test Pattern, Right Eye**Figure A-18 Full Field 135 Test Pattern, Right Eye**Figure A-19 Full Field 246 Test Pattern, Right Eye*

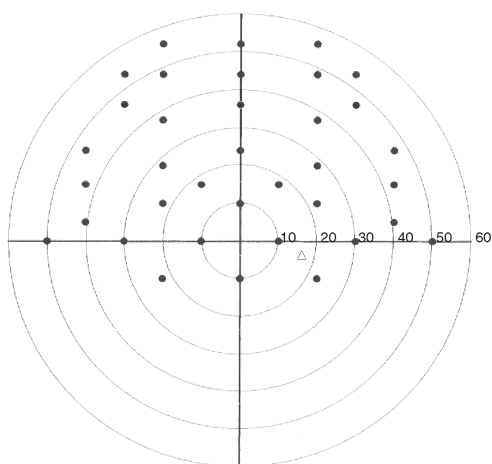


Figure A-20 Superior 36 Test Pattern, Right Eye

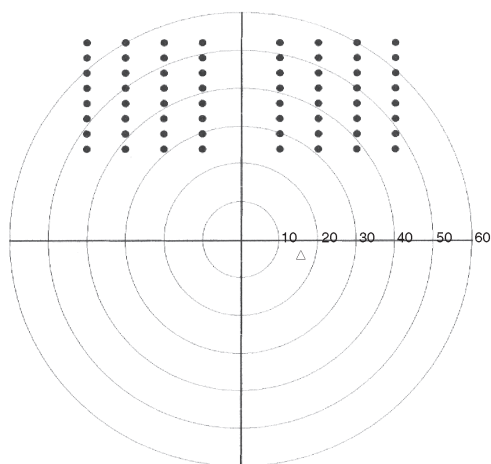


Figure A-21 Superior 64 Test Pattern, Right Eye

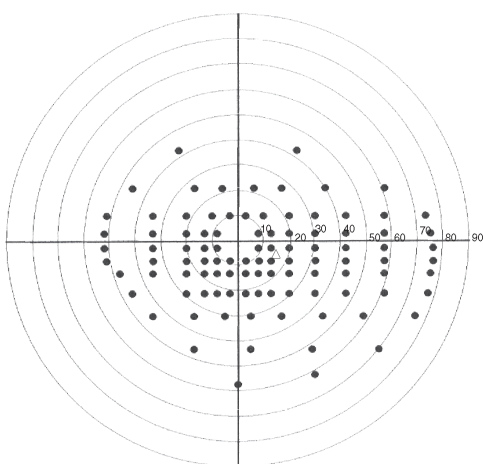


Figure A-22 Esterman Monocular Test Pattern, Right Eye

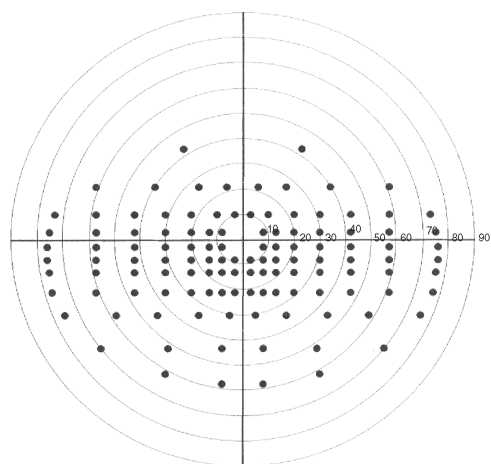


Figure A-23 Esterman Binocular Test Pattern

Test Parameters

Threshold Test Settings

The following table displays parameter settings available only when Static Threshold Testing is chosen. Default values are shown in bold.

Parameter	Parameter Settings	Description
Test Strategy	SITA-Faster	SITA Faster reduces test time approximately 30% faster than SITA Fast and 50% faster than SITA Standard for the same patient tests. It is available with the Central 24-2 pattern only. SITA Faster allows to add selected central locations back to the 24-2 test pattern to create an enhanced test pattern that can complete, on average, in the same time (or less) as a legacy 24-2 SITA Fast visual field. Resulting in a choice between running a quicker SITA Faster 24-2 visual field or a SITA Faster enhanced test pattern with the additional points. The Combined test pattern (24-2C) will be run using SITA Faster test strategy only.
	SITA-Fast	This is a faster version of SITA. SITA-Fast cuts testing time in half relative to the FastPac testing strategy, and is available for use with the Central 30-2, Central 24-2, Central 10-2, and Peripheral 60-4 test patterns.
	SITA-Standard	SITA-Standard cuts testing time in half relative to the Full Threshold strategy, and is available for use with the Central 30-2, Central 24-2, Central 10-2, and Peripheral 60-4 test patterns.
	SITA-SWAP (Models 850 and 860)	This strategy uses SITA with Blue-Yellow testing (see page A-9). SITA-SWAP testing is only available for use with the Central 24-2 test pattern.
	Full Threshold	The test strategy used prior to the adoption of SITA. Use with all Threshold test patterns. In Full Threshold testing, an initial stimulus is presented at a level the patient is expected to see. If seen, the stimulus intensity is decreased in 4 dB steps (0.4 log units) until the patient no longer sees the stimulus; if not seen, it is increased in 4 dB steps until seen. The instrument then changes direction, moving in 2 dB steps until a change in patient response occurs. The last stimulus seen by the patient is recognized as the threshold for that point.
	FastPac	FastPac is similar to the Full Threshold strategy, but decreases test time by about 40%. Use with all Threshold test patterns. It follows a similar stair-stepping technique as in Full Threshold, but uses 3 dB increments instead of 4 dB and crosses the threshold only once.
Foveal Threshold (Models 840, 850, and 860)	Off	A threshold value for the fovea will not be measured.
	On	A threshold value for the fovea will be determined at the beginning of the test. The foveal threshold test presents stimuli inside the Small Diamond fixation target; the Small Diamond will automatically be illuminated.
Fluctuation	Off	Threshold values for preselected points will not be determined twice. Some points may be retested even if fluctuation is turned off.
	On	Threshold values for ten (10) preselected points are retested to determine the variability of the patient's responses. Threshold values for the retested points are printed on the numeric printout and appear in parentheses directly below the first test result. Fluctuation values which differ significantly from normal are flagged with appropriate "p" (probability) values. Available for Full Threshold and FastPac tests only.

Table A-3

Suprathreshold Test Settings

The following table displays parameter settings available only when Suprathreshold Testing is chosen. Default values are shown in bold.

Parameter	Parameter Settings	Description
Test Strategy	Two Zone	For each point in the test pattern, a stimulus is presented 6 dB brighter than the expected threshold. Since testing is done with an intensity 6 dB brighter than the expected threshold, Points Not Seen are known to be at least 6 dB deep. Use with all Suprathreshold test patterns. Esterman tests use only the Two Zone test strategy.
	Three Zone	Same as Two Zone, except each missed point is measured again at a maximum intensity of 10,000 apostilbs (0 dB) to determine if the defect is absolute. Use with all Suprathreshold test patterns except the Esterman tests.
	Quantify Defects	Same as Two Zone, except the sensitivity at each missed point is measured relative to the expected threshold. Use with all Suprathreshold test patterns except the Esterman tests.
Test Mode	Age Corrected	Vision is most acute at the fovea and decreases toward the periphery of the retina resulting in a hill of vision with the peak occurring at the fovea. A hill of vision is assigned to the patient based on the patient's age. Age Corrected mode is used only with Size III, White stimuli, and can be used with all Suprathreshold test patterns except the Esterman tests.
	Threshold Related	A hill of vision is assigned only after threshold values for 4 primary points are determined. Use with all Suprathreshold test patterns except the Esterman tests.
	Single Intensity	Available only with the Two Zone test strategy. Esterman tests use only the Single Intensity test mode. The HFA uses a default intensity level of 10 dB to test the entire visual field. The default can be changed except with the Esterman tests.

Table A-4

Shared Settings

The following table displays parameter settings available for both Static Threshold and Suprathreshold testing. Default values are shown in bold.

Parameter	Parameter Settings	Description
Blue-Yellow/Short-Wave length Automated Perimetry or SWAP (Models 850 and 860)	Off/On	Blue-Yellow testing uses a blue stimulus presented on a yellow background.
Stimulus Color	White (all models)	White stimulus projected onto white bowl background. White is used only for the SITA-Standard, SITA-Fast, and Suprathreshold Age Corrected test strategies, as well as the Esterman test patterns.
	Red (Models 840, 850, and 860)	Red stimulus projected onto white bowl background.
	Blue (Models 850 and 860)	Blue stimulus projected onto white bowl background. Background is yellow, if Blue-Yellow is On .
Stimulus Size	I, II, III , IV, V	Five stimulus sizes (diameters) are available, ranging from Size I (smallest) to Size V (largest). SITA-Standard, SITA-Fast, and Suprathreshold Age Corrected test strategies, as well as the Esterman test patterns use only Size III. SITA-SWAP uses only Size V.
Fixation Monitoring	Gaze/Blind Spot	The Blind Spot and Gaze Tracking system are both activated.

Table A-5

Parameter	Parameter Settings	Description
	Gaze Tracking (Models 840, 850, and 860)	The Gaze Tracking system automatically measures gaze direction at the time of stimulus presentation. Note: Gaze Tracking is the default for SITA Faster.
	Blind Spot	The test program periodically presents a stimulus in the patient's blind spot. If the patient is fixating well, he or she should not see the blind spot check stimulus. The blind spot check stimulus always matches the test stimulus size.
	Off	Disables all fixation monitoring. The operator should monitor fixation with the video eye monitor.
Fixation Target	Central	Yellow light in the center of the bowl. This is the only option available for the Threshold Peripheral 60-4, and the Suprathreshold Esterman tests.
	Small Diamond	The Small Diamond is located below the Central target, and should be used when a patient cannot see the central fixation light (e.g. macular degeneration). The patient should look in the center of the diamond formed by the four lights.
	Large Diamond	The Large Diamond is located below the Central target and is useful for patients with central scotoma who cannot see either the Central fixation light or the Small Diamond.
	Bottom LED	Some tests have points in the superior visual field that require a lower fixation light than the central target. The target used is the Bottom LED of the Large Diamond. When testing with the Superior 64 or Superior 36 Suprathreshold tests, the Bottom LED is the default fixation target. It is illuminated automatically at the start of a test.
Test Speed	Normal	Two stimulus presentation speeds are available.
	Slow	The test speed may be changed while a test is in progress to allow the patient some additional time to respond.

Table A-5